

3-Mark Questions & Answers -- Data Structures Module 4: Trees

Q1. Explain the three Depth-First Traversals of a Binary Tree with examples.

1. Preorder (Root -> Left -> Right): Visit root first, then left and right subtrees.

2. In

Q2. Explain the four AVL Tree rotations with diagrams.

1. LL Rotation: Single Right Rotation at unbalanced node z (Left-Left insertion).

2. RR

Q3. Explain the steps of Huffman Encoding algorithm.

1. Count character frequencies and insert each into a Min-Heap.

2. Re

Q4. State the 5 properties of Red-Black Trees.

1. Every node is either RED or BLACK.

2. Th